

## ESTEVAN POINT, BC, Canada

WMO: 718940

Lat: 49.3833N

Lon: 126.5431W

Elev: 6

StdP: 101.26

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -1.7       | -0.6 | -6.5                          | 2.2 | 0.4  | -3.8 | 2.8 | 1.0  | 19.3                  | 7.5  | 17.5 | 7.7  | 3.3                   | 60   | 0.695 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.0                    | 19.8            | 16.4 | 18.6 | 15.8 | 17.6 | 15.2 | 17.0                | 19.0 | 16.2 | 18.0 | 15.7 | 17.2 | 4.7                  | 320  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 16.1                            | 11.4 | 17.7 | 15.5 | 11.0 | 16.9 | 15.0 | 10.6 | 16.3 | 47.5          | 19.2 | 45.4 | 18.1 | 43.8 | 17.3 | 20.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 15.8          | 13.5 | 11.9 | -3.8                       | 23.4 | 1.6                | 2.1 | -5.0                                               | 24.9 | -5.9       | 26.2 | -6.8       | 27.3 | -8.0       | 28.9 |    |    |
| (5)               |      |      | -4.8                       | 18.2 | 1.8                | 0.8 | -6.0                                               | 18.8 | -7.1       | 19.3 | -8.1       | 19.8 | -9.3       | 20.4 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 9.8         | 6.0   | 5.8   | 6.5   | 8.2   | 10.8  | 12.8  | 14.5  | 14.8  | 13.6  | 10.4  | 7.6   | 5.7   |
|                                                                        | DBStd       | 3.95        | 2.57  | 2.24  | 2.04  | 1.85  | 1.88  | 1.52  | 1.43  | 1.33  | 1.66  | 1.88  | 2.61  | 2.57  |
|                                                                        | HDD10.0     | 655         | 126   | 118   | 108   | 59    | 13    | 1     | 0     | 0     | 0     | 17    | 78    | 135   |
|                                                                        | HDD18.3     | 3132        | 383   | 351   | 366   | 304   | 233   | 165   | 118   | 111   | 142   | 245   | 322   | 393   |
|                                                                        | CDD10.0     | 565         | 1     | 0     | 1     | 5     | 38    | 86    | 141   | 148   | 109   | 31    | 6     | 1     |
|                                                                        | CDD18.3     | 1           | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDH23.3     | 3           | 0     | 0     | 0     | 0     | 0     | 1     | 1     | 1     | 1     | 0     | 0     | 0     |
|                                                                        | CDH26.7     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 5.1         | 5.3   | 5.1   | 5.3   | 5.4   | 5.2   | 5.1   | 5.0   | 4.3   | 4.4   | 4.8   | 5.6   | 5.5   |
| (15) Precipitation                                                     | PrecAvg     | 3276        | 418   | 356   | 331   | 259   | 163   | 136   | 82    | 104   | 171   | 386   | 446   | 418   |
|                                                                        | PrecMax     | 4037        | 706   | 735   | 559   | 505   | 370   | 288   | 349   | 276   | 373   | 796   | 736   | 784   |
|                                                                        | PrecMin     | 2305        | 97    | 134   | 50    | 100   | 50    | 19    | 6     | 6     | 14    | 84    | 158   | 207   |
|                                                                        | PrecStd     | 428         | 173   | 143   | 110   | 109   | 86    | 73    | 73    | 73    | 88    | 176   | 133   | 140   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 12.1  | 11.8  | 13.6  | 15.6  | 18.9  | 21.1  | 21.7  | 21.2  | 20.9  | 16.4  | 13.8  | 12.2  |
|                                                                        |             | MCWB        | 8.5   | 9.3   | 9.7   | 12.5  | 13.8  | 16.0  | 17.4  | 17.1  | 17.1  | 13.3  | 12.2  | 9.2   |
|                                                                        | 2%          | DB          | 10.6  | 10.6  | 11.6  | 13.7  | 16.3  | 17.7  | 19.2  | 19.5  | 18.7  | 15.1  | 12.6  | 10.8  |
|                                                                        |             | MCWB        | 9.5   | 9.0   | 9.1   | 10.7  | 13.3  | 14.7  | 16.1  | 16.5  | 16.0  | 13.3  | 11.4  | 9.6   |
|                                                                        | 5%          | DB          | 9.9   | 9.9   | 10.6  | 12.4  | 14.9  | 16.5  | 18.2  | 18.3  | 17.4  | 14.1  | 11.9  | 10.1  |
|                                                                        |             | MCWB        | 9.3   | 8.6   | 9.0   | 10.0  | 12.4  | 14.0  | 15.5  | 16.0  | 15.3  | 12.8  | 11.0  | 9.2   |
|                                                                        | 10%         | DB          | 9.3   | 9.1   | 9.7   | 11.4  | 13.9  | 15.5  | 17.2  | 17.3  | 16.4  | 13.3  | 11.3  | 9.3   |
|                                                                        |             | MCWB        | 8.8   | 8.2   | 8.5   | 9.5   | 11.7  | 13.4  | 15.0  | 15.4  | 14.7  | 12.3  | 10.5  | 8.6   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 10.6  | 10.7  | 11.0  | 12.9  | 14.6  | 16.5  | 18.0  | 17.7  | 17.8  | 15.0  | 13.2  | 11.0  |
|                                                                        |             | MCDB        | 10.7  | 11.0  | 11.9  | 15.3  | 17.2  | 20.4  | 21.0  | 19.8  | 19.8  | 15.5  | 13.4  | 11.2  |
|                                                                        | 2%          | WB          | 10.0  | 9.9   | 10.2  | 11.6  | 13.7  | 15.2  | 16.7  | 17.0  | 16.5  | 14.0  | 12.0  | 10.1  |
|                                                                        |             | MCDB        | 10.3  | 10.3  | 11.1  | 13.2  | 15.9  | 17.7  | 18.5  | 19.0  | 18.1  | 14.4  | 12.4  | 10.3  |
|                                                                        | 5%          | WB          | 9.5   | 9.2   | 9.5   | 10.6  | 12.8  | 14.4  | 15.9  | 16.3  | 15.6  | 13.2  | 11.2  | 9.2   |
|                                                                        |             | MCDB        | 9.8   | 9.7   | 10.3  | 11.8  | 14.7  | 16.3  | 17.6  | 18.0  | 16.9  | 13.7  | 11.6  | 9.6   |
|                                                                        | 10%         | WB          | 9.0   | 8.5   | 8.8   | 9.9   | 12.1  | 13.7  | 15.3  | 15.7  | 15.0  | 12.5  | 10.5  | 8.7   |
|                                                                        |             | MCDB        | 9.3   | 9.1   | 9.6   | 11.0  | 13.6  | 15.2  | 16.8  | 17.1  | 16.0  | 13.1  | 11.0  | 9.1   |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 4.2   | 5.1   | 5.2   | 5.7   | 5.5   | 4.9   | 4.9   | 5.0   | 5.2   | 4.9   | 4.8   | 4.3   |
|                                                                        |             | MCDBR       | 3.9   | 5.7   | 6.6   | 7.5   | 7.1   | 6.6   | 6.4   | 6.6   | 7.0   | 5.8   | 4.8   | 4.3   |
|                                                                        | 5% WB       | MCWBR       | 3.0   | 4.2   | 4.7   | 5.1   | 4.8   | 4.3   | 4.0   | 4.2   | 4.7   | 4.2   | 4.1   | 3.6   |
|                                                                        |             | MCDWR       | 3.0   | 4.2   | 5.2   | 6.3   | 6.6   | 6.1   | 6.0   | 6.1   | 6.1   | 4.5   | 4.2   | 3.8   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.323 | 0.325 | 0.349 | 0.371 | 0.384 | 0.363 | 0.352 | 0.356 | 0.349 | 0.351 | 0.330 | 0.311 |
|                                                                        |             | taud        | 2.455 | 2.509 | 2.453 | 2.387 | 2.377 | 2.467 | 2.544 | 2.542 | 2.545 | 2.479 | 2.454 | 2.466 |
|                                                                        | Ebn at Noon |             | 731   | 829   | 863   | 877   | 876   | 896   | 902   | 884   | 855   | 782   | 712   | 693   |
|                                                                        |             | Edh at Noon | 63    | 75    | 93    | 109   | 115   | 106   | 97    | 93    | 84    | 76    | 62    | 55    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 0.83  | 1.75  | 2.70  | 4.24  | 5.38  | 5.81  | 5.93  | 5.16  | 3.75  | 2.04  | 1.00  | 0.68  |
|                                                                        | RadStd      |             | 0.10  | 0.27  | 0.32  | 0.37  | 0.53  | 0.54  | 0.56  | 0.47  | 0.29  | 0.27  | 0.17  | 0.10  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            |       | N/A     | N/A    | N/A     | -0.95 | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (19 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page